



Tribhuvan University
Faculty of Humanities and Social Sciences

A PROJECT REPORT ON
‘NCE’ – Nepal Freelance

Submitted to
Department of Computer Application
Samarpan Academy

In partial fulfillment of the requirements for the Bachelors in Computer Application

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August 2024

Under the Supervision of
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Tribhuvan University
Faculty of Humanities and Social Sciences
Samarpan Academy

Supervisor's Recommendation

I hereby recommend that this project prepared under my supervision by **Aryan Tamang** and **Sanjana Prajapati** entitled 'NCE' – **Nepal Freelance** in partial fulfillment of the requirement for the degree of Bachelor of Computer Application is recommended for the final evaluation.

.....

SIGNATURE

Suman Dhital

Supervisor

Samarpan Academy



Tribhuvan University
Faculty of Humanities and Social Sciences
Samarpan Academy

LETTER OF APPROVAL

This is to certify that this project is prepared by **Aryan Tamang** and **Sanjana Prajapati** entitled '**NCE**' – **Nepal Freelance** in partial fulfillment of the requirements for the degree of Bachelors in Computer Application has been evaluated. In our opinion it is satisfactory in terms of scope and quality as a project for the required degree.

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ACKNOWLEDGEMENT

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Aryan Tamang

Sanjana Prajapati

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List of Abbreviations

CASE tools – Computer Aided Software Engineering tools

DFD – Data Flow Diagram

ER – Entity Relationship

FRs – Functional Requirements

NFRs – Non-Functional Requirements

SQL – Structured Query Language

UI – User Interface

ABSTRACT

In today's rapidly evolving digital landscape, there is a growing demand for an efficient and accessible freelancing platform that connects companies with skilled freelancers. This project presents the development of a freelancing platform designed to bridge this gap by allowing companies to post jobs and freelancers to apply for them seamlessly. Key functionalities include user registration, job posting, proposal submission, real-time chat, and secure payments. The Waterfall Model was adopted as the software development methodology, following a structured approach through requirement analysis, system design, implementation, testing, and deployment. The frontend is developed using Next.js, ensuring a fast, dynamic, and responsive user interface. The backend is powered by Django Ninja, a modern Python framework ideal for building robust RESTful APIs. PostgreSQL serves as the database, securely managing data related to users, jobs, proposals, and transactions. The system defines clear roles for companies, freelancers, and admins, ensuring smooth workflow and usability. This platform aims to simplify freelance work management and contribute to the growing digital economy in Nepal.

Keywords: *Freelancing Platform, Waterfall Model, Next.js, Django Ninja, PostgreSQL, Job Posting, Proposal System, Company-Freelancer Communication, Digital Economy Nepal.*

Chapter 1: Introduction

1.1 Introduction

Freelancing means working online for different people or companies without going to an office. People with skills like graphic design, writing, programming, video editing, and more can earn money from home using just a mobile or computer. In Nepal, many young people want to do freelancing because it gives them freedom and a chance to earn well. It's also helpful for students, women at home, and people in rural areas.

However, most Nepali freelancers use international websites like Upwork, Fiverr, or Freelancer.com. These platforms are popular, but they are not made for Nepali users. Freelancers have to compete with people from all over the world, and that makes it hard to get jobs. Also, these websites use foreign payment systems like PayPal or Payoneer, which are not easy to use in Nepal. This creates problems when freelancers want to receive their payments.

Because of this, there is a need for a freelancing platform that is made specially for Nepal. A platform where Nepali freelancers and companies can connect, work together, and send or receive payments easily using local system like Khalti.

This platform will be developed using Django (a web framework) for the backend and a mobile app for easy access on phones. It will be secure, fast, and designed to match the needs of Nepali users.

With this system, Nepali freelancers will have a better chance to grow. It will also help small companies find skilled people easily. This project will support digital growth in Nepal and give more opportunities to people who want to work online.

1.2 Problem Statement

Nepal currently does not have a dedicated freelancing platform that is specially designed for local users. Most Nepali freelancers rely on international websites like Upwork and Fiverr to find work. While these platforms are useful, they are not well-suited for the specific needs of people in Nepal. The competition is very high on those global platforms because freelancers from all over the world are bidding on the same projects. As a result, many talented Nepali freelancers struggle to get their first job or earn enough money through those sites.

Another big issue is the payment system. International platforms mostly use foreign payment methods like PayPal and Payoneer, which are either not available or difficult to access in Nepal. This makes it hard for freelancers to receive the money they earn. Similarly, Nepali companies who want to hire local freelancers face difficulties using these platforms, as they have to deal with international currencies and processes.

This shows that there is a clear gap in the market for a freelancing platform made for Nepal. Freelancers and companies in Nepal need a simple, user-friendly system that supports local languages, works well with Nepali payment systems like eSewa and Khalti, and focuses on the local market. A platform like this would help more people in Nepal earn online, support small businesses, and boost the country's digital economy.

1.3 Objectives

- To create a freelancing website for Nepali users.
- To make a safe and secure platform with local payment options (like eSewa and Khalti).
- To provide equal opportunities for Nepalese freelancers to showcase their skills

1.4 Scopes and Limitations

Scope

- Freelancers can show their skills and get jobs.
- Companies can post jobs and hire people from anywhere.
- Payments are handled safely through the platform.
- Allow users to register as freelancers or companies.
- Support chatting between freelancers and companies.

Limitations

- Needs internet – no internet, no access.
- If the website crashes, users can't do anything until it's fixed.
- Arguing over work quality or payment can happen sometimes.
- In the first version, only basic features will be available.
- No video calling or real-time voice chat in the beginning.

1.5 Report Organization

This project report is divided into six chapters, each explaining a specific part of the freelancing platform development:

- **Chapter 1: Introduction:** This chapter explains the main idea behind the project. It describes the purpose of the freelancing platform, the problems it aims to solve in Nepal, its goals, scope, and the limitations of the current version.
- **Chapter 2: Background Study and Literature Review:** This chapter looks at existing freelancing platforms like Upwork and Fiverr. It highlights their features, advantages, and the common problems faced by Nepali users. It also explains why a local solution is needed.
- **Chapter 3: System Analysis and Design:** This chapter discusses the methods used in developing the system. It includes the system requirements (both functional and non-functional), feasibility study, ER diagram, data flow diagram (DFD), system architecture, database design, user interface layout, and how the payment system will work.
- **Chapter 4: Implementation and Testing:** This chapter describes how the system was developed using tools like Django and mobile programming frameworks. It

also explains each module of the system and includes test cases for unit testing and system testing to ensure everything works properly.

- **Chapter 5: Conclusion and Future Recommendations:** This chapter summarizes the entire project, the challenges faced during development, and how they were solved. It also gives suggestions for future updates and improvements to make the platform more useful and user-friendly.

Chapter 2: Background Study and Literature Review

2.1 Background Study

Freelancing has become a significant employment model globally, allowing individuals to offer specialized services remotely and earn income through digital platforms. Freelancing platforms like Upwork, Fiverr, and Freelancer.com provide structured environments where freelancers and clients can connect, collaborate, and process payments securely [1]. These platforms support a wide range of services such as writing, design, programming, and marketing.

However, freelancers in Nepal often face challenges when engaging with these global platforms. Due to intense international competition, new users struggle to secure their first projects. Additionally, most platforms rely on payment methods like PayPal, which are either unsupported or difficult to use in Nepal [2]. The high commission rates and foreign currency conversions also reduce the net income of Nepali freelancers. On the client side, international platforms can be complicated to use due to unfamiliar payment systems and the lack of local language support.

Given these limitations, there is a clear need for a freelancing platform specifically designed for Nepal. A localized solution could support payment gateways like eSewa and Khalti, provide Nepali language support, and reduce competition by focusing on the local market [3]. Such a platform would help Nepali freelancers and companies collaborate more effectively and would contribute to the growth of the country's digital economy.

2.2 Literature Review

Global freelancing platforms have revolutionized remote work. Upwork allows clients to post projects and freelancers to submit proposals, offering features like milestone payments, contract security, and a rating system [1]. However, its competitive nature and strict profile review process make it difficult for new users to establish themselves [4].

Fiverr, on the other hand, operates on a gig-based model where freelancers post services that clients can purchase directly. While easier to start, Fiverr's reliance on ratings and high service fees (20%) can be a disadvantage to new and lower-income freelancers [5]. Freelancer.com also allows bidding on projects but faces issues like spam bids, unclear job scopes, and frequent disputes [6].

A recent study by ICT for Development Nepal found that over 60% of Nepali youth are interested in freelancing but cite challenges such as a lack of local payment options, language barriers, and insufficient exposure [7]. Current local platforms like MeroJob and Kumari Job primarily serve full-time employment needs and do not offer freelance job functionalities, highlighting a gap in the market [8].

Research supports that localized platforms can help bridge this gap. They can provide job access in the native language, support mobile-first users, integrate local wallets, and reduce commission fees. As internet access increases in Nepal, the demand for digital work platforms will continue to grow, making a Nepal-focused freelancing site both necessary and viable [9].

Chapter 3: System Analysis and Design

3.1 System Analysis

A system development methodology in software engineering serves as a guide for planning, creating, and managing the process of building an information system. Some commonly used development approaches include the Waterfall model, Prototyping, Spiral model, and Extreme Programming, among others. Since this project focuses on the design and development of a software application, particularly a web-based system, it is important to refer to and consider appropriate software development models that support the system's development and deployment process.

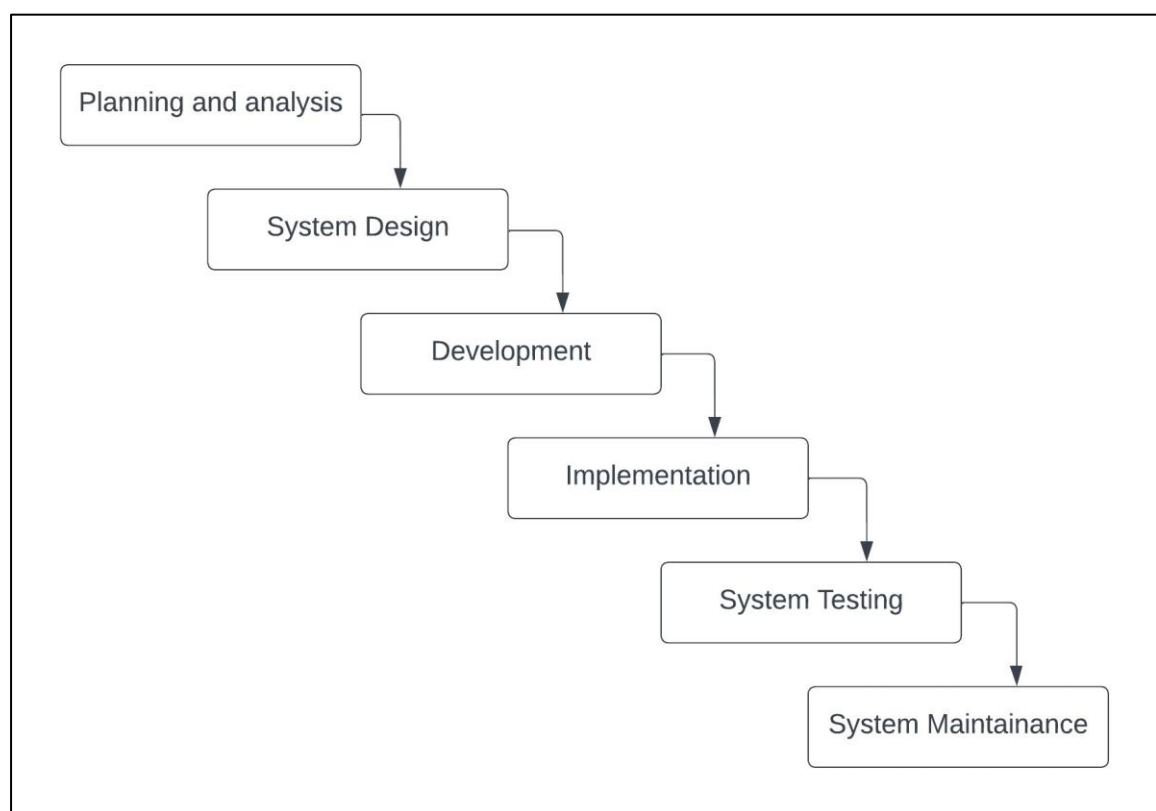


Figure 1: Waterfall Model

This system has well defined requirements and functionalities, specific documentation, ample time and well understood technology. So, to build the system, we have used waterfall methodology.

The Waterfall Model was the first Process Model to be introduced. It is also referred to as a linear-sequential life cycle model. It is very simple to understand and use. In a waterfall

model, each phase must be completed before the next phase can begin and there is no overlapping in the phases. It is easy to arrange tasks and clearly defined stages.

3.1.1 Requirement Analysis

a) Functional Requirements

The system includes three main roles: **Admin**, **Company**, and **Freelancer**. The main features needed during the development of this freelancing platform are as follows:

i. Admin Module

- Admin can log in and log out of the system
- Admin can create, edit, or delete freelancer and company accounts
- Admin can manage job categories and skills
- Admin can view reports related to projects and payments
- Admin can manage the platform's content, such as featured jobs and users

ii. Company Module

- Companies can register and log in to the system
- Companies can post new freelance jobs/projects
- Companies can browse freelancer profiles and hire them
- Company can make payments through local method (Khalti)

iii. Freelancer Module

- Freelancer can register and log in to the system
- Freelancer can create and manage their profile
- Freelancer can browse and apply for posted jobs
- Freelancer can communicate with companies through the platform
- Freelancer can receive payments via supported gateways

b) Non-functional Requirements

The non-functional requirements of the NCE focus on how the system performs rather than what it does. These include **Efficiency**, **Reliability**, **Usability**, and **Implementation**.

- **Efficiency**

The platform is designed to be efficient in handling multiple users simultaneously,

including freelancers and companies. With just a basic internet connection, users can browse jobs, submit proposals, and communicate in real-time. The system ensures quick load times and fast response rates, providing a smooth user experience.

- **Reliability**

The system ensures that user data such as profiles, job listings, proposals, and payments is securely handled and consistently available. Key features like registration, login, job posting, and proposal submission are designed to work accurately without failure, always ensuring system reliability.

- **Usability**

The platform is built with user-friendliness in mind, targeting both freelancers and companies, including those who may not be highly tech-savvy. The interface is intuitive and easy to navigate, allowing users to post jobs, manage profiles, communicate, and handle transactions effortlessly.

- **Implementation**

The platform's front-end is developed using Mobile Application, providing a dynamic and responsive user interface. The backend is powered by Django Ninja enabling scalable and efficient server-side operations. PostgreSQL is used for database management, offering flexible and efficient data storage and retrieval, well-suited for user-driven content like job listings and messages.

3.1.2 Use Case Diagram

In this project, a **Use Case Diagram** is used to show how different users interact with the system. It helps to visually understand what actions are performed by each role. The main actors in our system are **Admin**, **Company**, and **Freelancer**.

The system boundary is shown as a rectangle, which includes all the functions of the platform. The “**Include**” relationships represent essential tasks, like login, that must be completed to access other features. “**Extend**” relationships show optional or conditional tasks that enhance user experience.

- The **admin** must log in first. After logging in, they can manage users, categories, and monitor reports.
- The **company** must register, then log in. After that, they can post projects, hire freelancers, and make payments.

- The **freelancer** must also register and log in. Once logged in, they can apply for jobs, communicate with companies, and receive payments.

This diagram clearly outlines how each role functions and interacts with the platform to ensure a smooth freelancing experience.

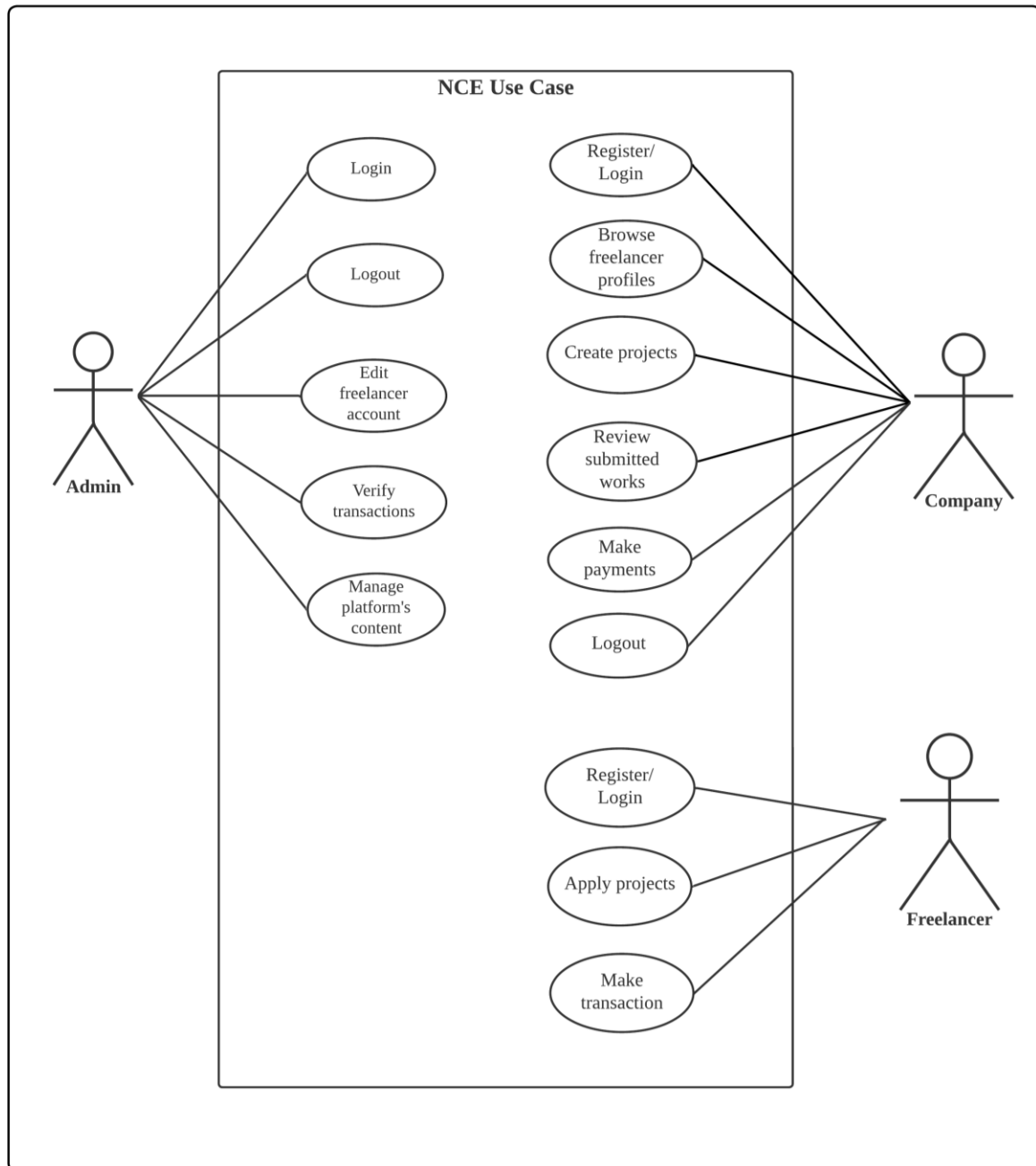


Figure 2: Use Case Diagram of Nepal Freelance

3.2 Feasibility Analysis

Feasibility analysis is the process of evaluating how a proposed system will impact an organization or target users. It helps determine whether the project is viable and worth pursuing. The analysis considers both positive and negative impacts, and a system is deemed feasible if the positives outweigh the negatives. In the context of the NCE, the feasibility study is conducted under the following categories:

- **Technical Feasibility**

Technical feasibility evaluates whether the required technology and resources are available and sufficient to build NCE. The project will be developed as a **mobile application**, with the front-end built using **Java** (Android), which is a mature and widely used programming language for mobile development. The back-end will be developed using **Django Ninja**, a modern, fast, and lightweight Python web framework ideal for building secure APIs efficiently. For data storage and management, the system will use **PostgreSQL**, a powerful and reliable open-source relational database system known for its robustness and advanced querying capabilities. All selected technologies are well-documented, scalable, and compatible, making the overall system technically feasible. Additionally, these tools have active communities and support resources, which can assist during development and maintenance.

- **Operational Feasibility**

Operational feasibility assesses how well the platform will function once implemented. NCE is designed to streamline interactions between freelancers and companies in Nepal. It supports essential operations like job posting, project tracking, communication, and payment handling. The system is user-friendly and can be adopted by people with basic computer and internet knowledge, making it operationally sound. Moreover, admin tools are integrated for platform moderation and user management, ensuring smooth ongoing operations.

- **Economic Feasibility**

Economic feasibility determines whether the project is financially viable. Developing NCE using open-source technologies significantly reduces costs. Initial investments may include development time, server hosting, domain registration, and marketing. However, the

potential for generating revenue through service fees, premium listings, or membership plans makes the platform economically feasible in the long run. As the demand for remote and freelance work is growing in Nepal, the platform has strong potential to become sustainable and profitable.

• Schedule Feasibility

Schedule feasibility analyzes whether the project can be completed within a reasonable time frame. Based on the scope and required features, the development of NCE is achievable within a typical academic or startup project timeline.

Gantt charts: Gantt chart is a bar chart that provides a visual view of tasks scheduled over time. A Gantt chart is used for planning projects of all sizes, and it is a useful way of showing what work is scheduled to be done on a specific day- It can also help you view the start and end dates of a project in one simple chart. In our project, we used Ms. Excel for developing the Gantt chart which is shown below in the figure.

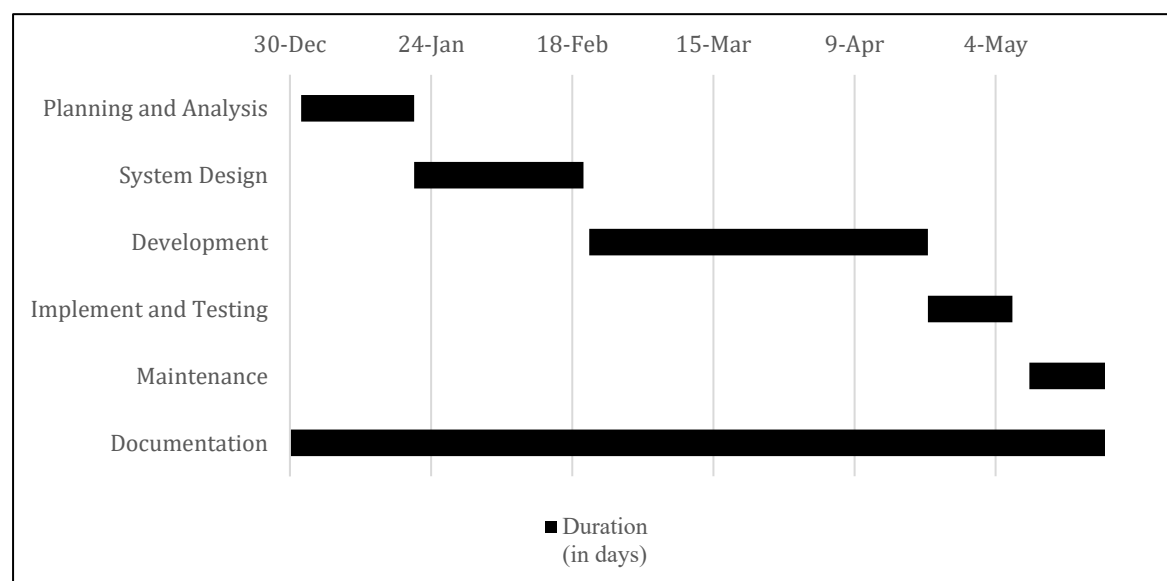


Figure 3: Gantt Chart of Nepal Freelance

3.2.1 Data Modeling (ER-Diagram)

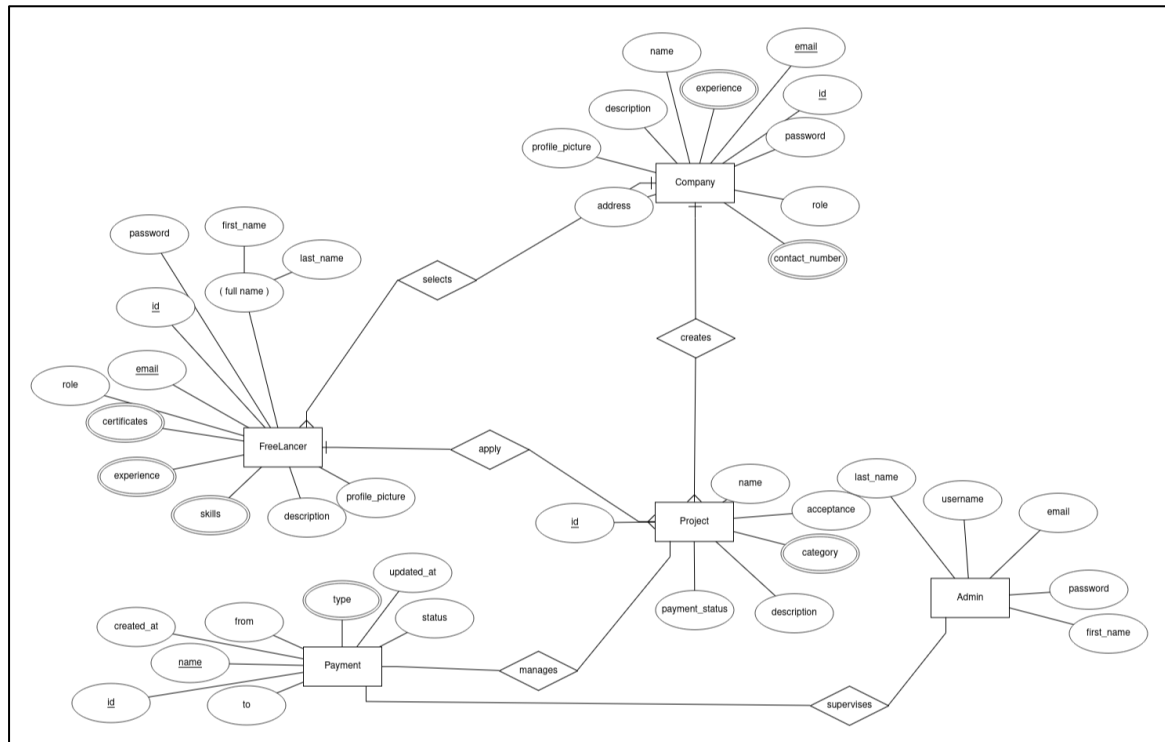


Figure 4: ER Diagram of Nepal Freelance

3.2.2 Process Modeling (DFD)

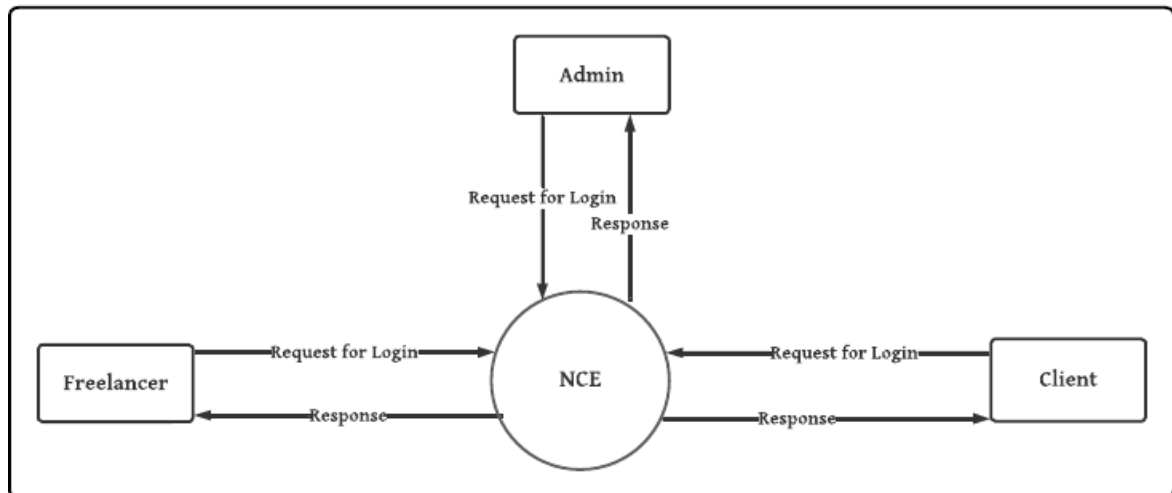


Figure 5: Context Diagram or Level 0 DFD of Nepal Freelance

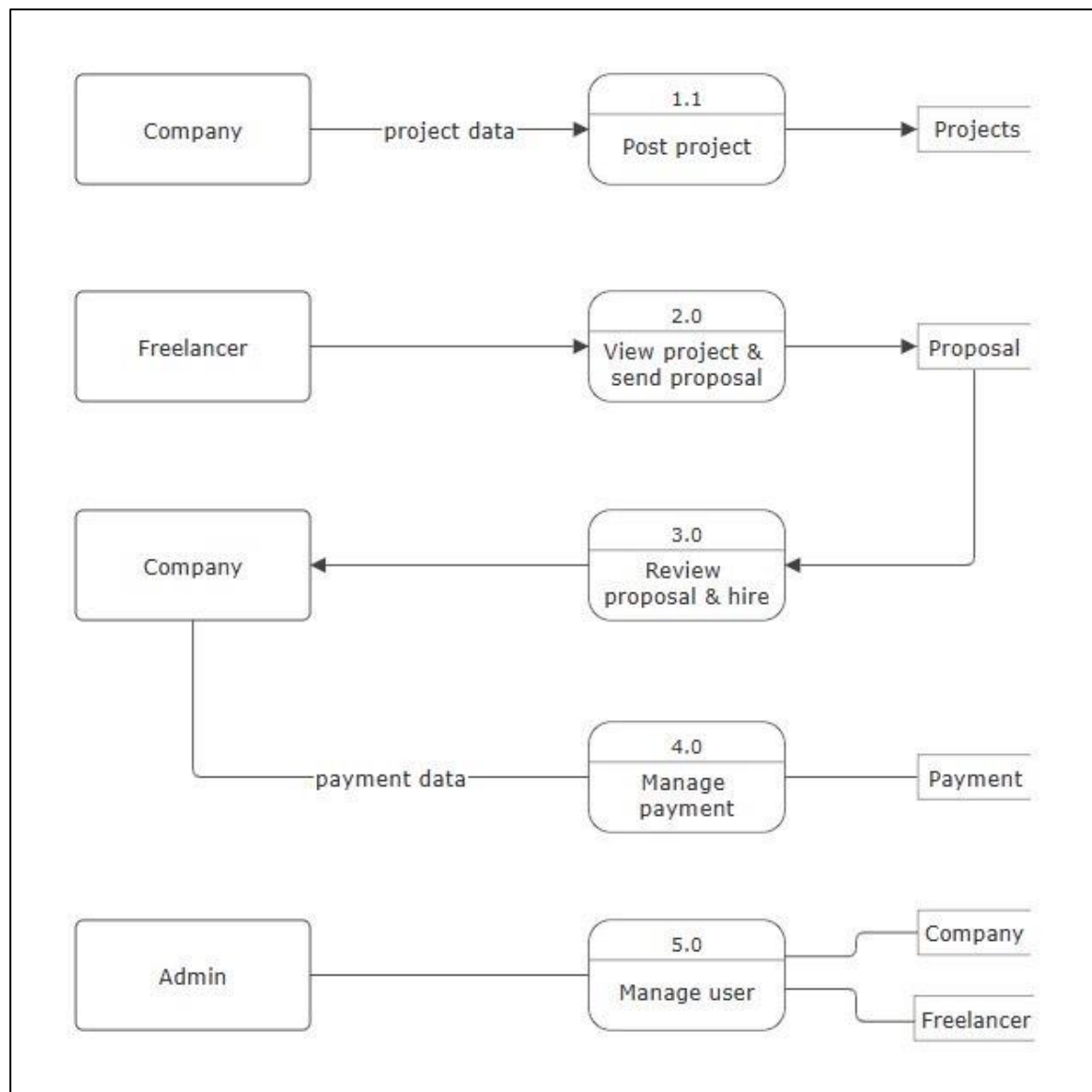


Figure 6: Level 1 DFD of Nepal Freelance

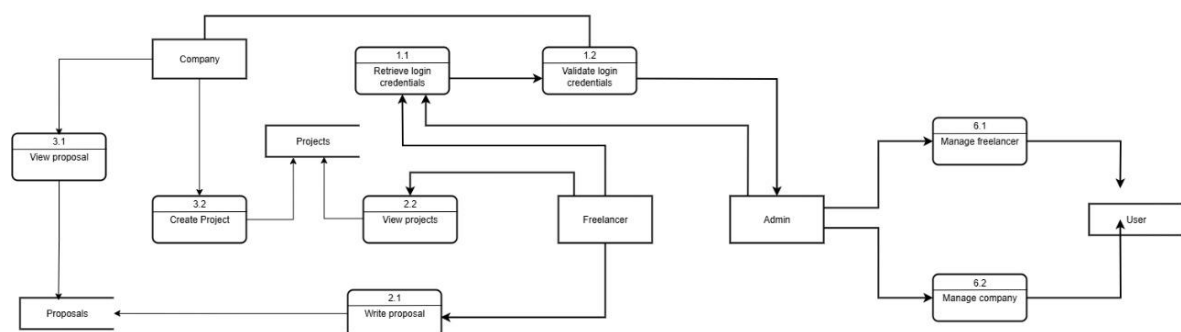


Figure 7: Level 2 DFD of Nepal Freelance

3.3 System Design

It will be vital to highlight and take into consideration several models used in software development and deployment given that this project involves the design and implementation of a software system, regardless of whether it is web-based. The process of building or changing a system, as well as the procedures, techniques, models, and development approaches, is referred to as system development.

3.3.1 Architecture Design:

Architectural design is a process for identifying the sub-systems making up a system and the framework for sub-system control and communication.

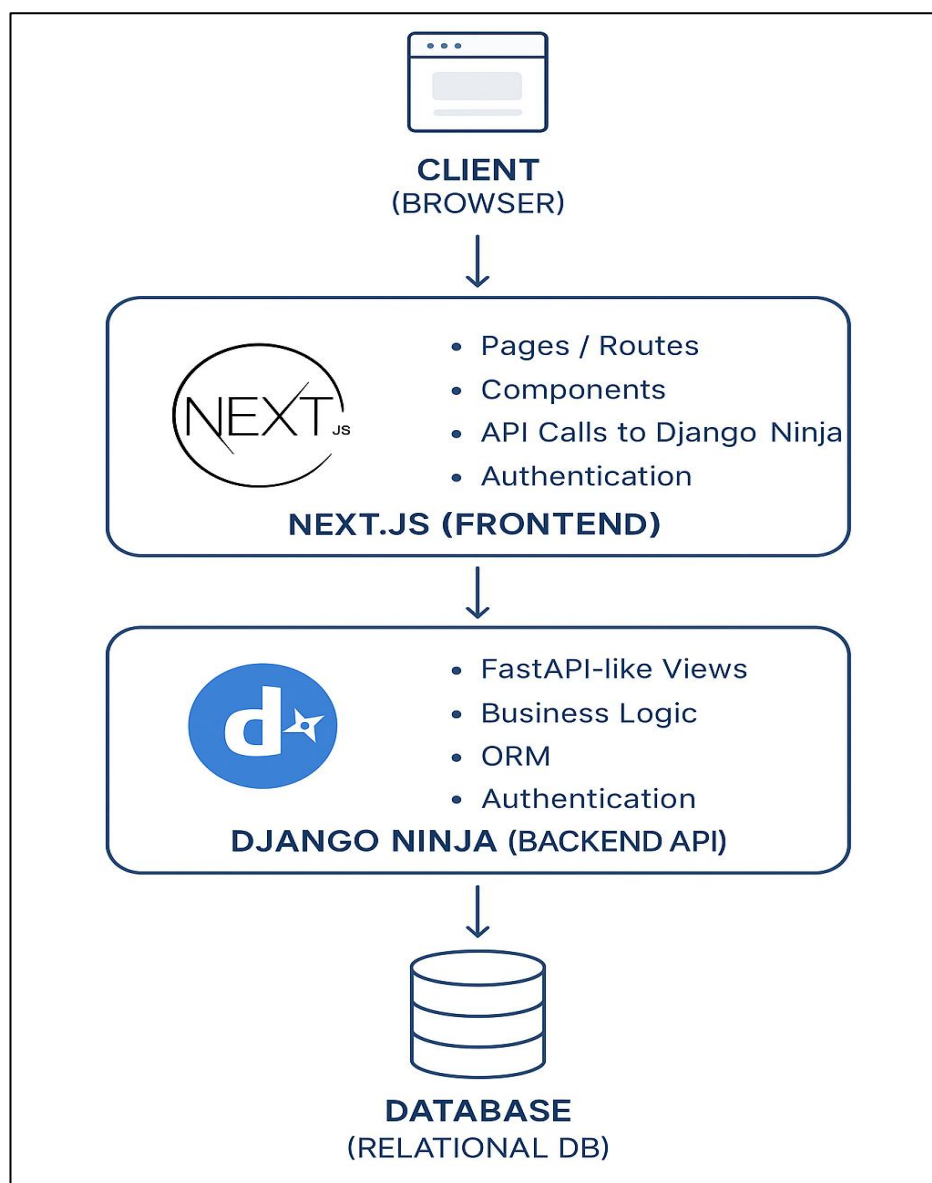


Figure 8: Architecture Design of Nepal Freelance

3.3.2 Interface Design (UI/UX)

The image shows a login form titled "Sign in" centered on a light gray background. The form is a white card with rounded corners. At the top, the title "Sign in" is in bold, followed by the instruction "Enter your credentials to access your account". Below this are three input fields: "Email" with a placeholder "Enter your email", "Password" with a placeholder "Enter your password" and a toggle icon (an eye in a circle), and "Role" with a placeholder "Select your role" and a dropdown arrow. A dark gray "Sign in" button is positioned below the fields. At the bottom of the card, it says "Don't have an account? [Sign up](#)".

Figure 9: Login Page

Create Account

Enter your information to create a new account

First Name

Enter your first name

Last Name

Enter your last name

Email

Enter your email

Password

Create a strong password

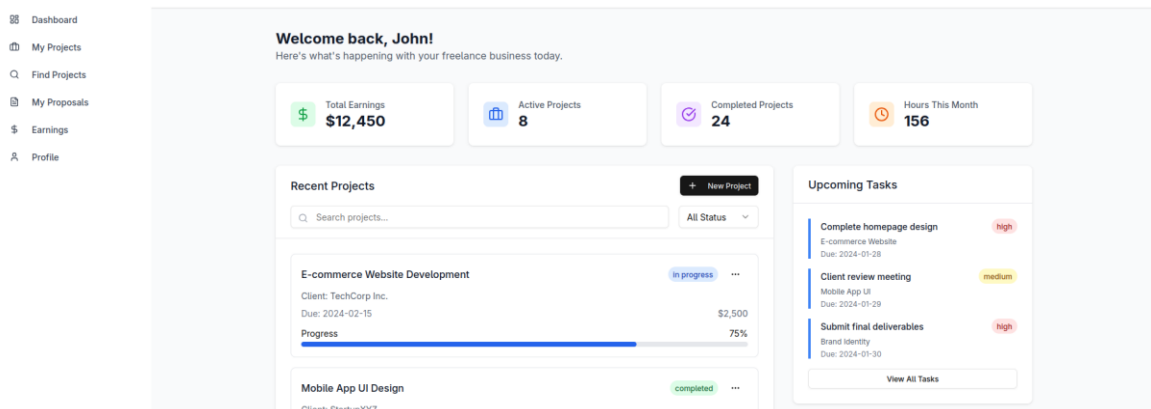
Role

Select your role

Create Account

Already have an account? [Sign in](#)

Figure 10: Register Page



My Projects

Track and manage your active projects

3 total projects

All Status

E-commerce Platform Development

In progress

Build a modern e-commerce platform with React and Node.js

Client: TechCorp Inc. \$ \$15,000 Due: 2024-03-15

Progress

65%

Project Milestones

✓ Project Setup	\$3,000	Paid
✓ Frontend Development	\$6,000	Paid
🕒 Backend API	\$4,000	
🕒 Testing & Deployment	\$2,000	

Message Client

View Details

Figure 12: My Projects Page

Find Projects

Discover new opportunities that match your skills

3 projects available

All Categories

All Budgets

Newest

React Native Mobile App Development

♥

We need an experienced React Native developer to build a cross-platform mobile app for our e-commerce business. The app should include user authentication, product catalog, shopping cart, and payment integration.

\$ \$8,000 - \$12,000

📅 2-3 months

📍 San Francisco, USA

React Native

JavaScript

API Integration

Mobile Development

T

TechStart Solutions

Rating: 4.8/5

Verified

12 proposals

Posted 2024-01-25

Experience: Intermediate

Category: Mobile Development

View Details

Send Proposal

Figure 13: Find Project Page

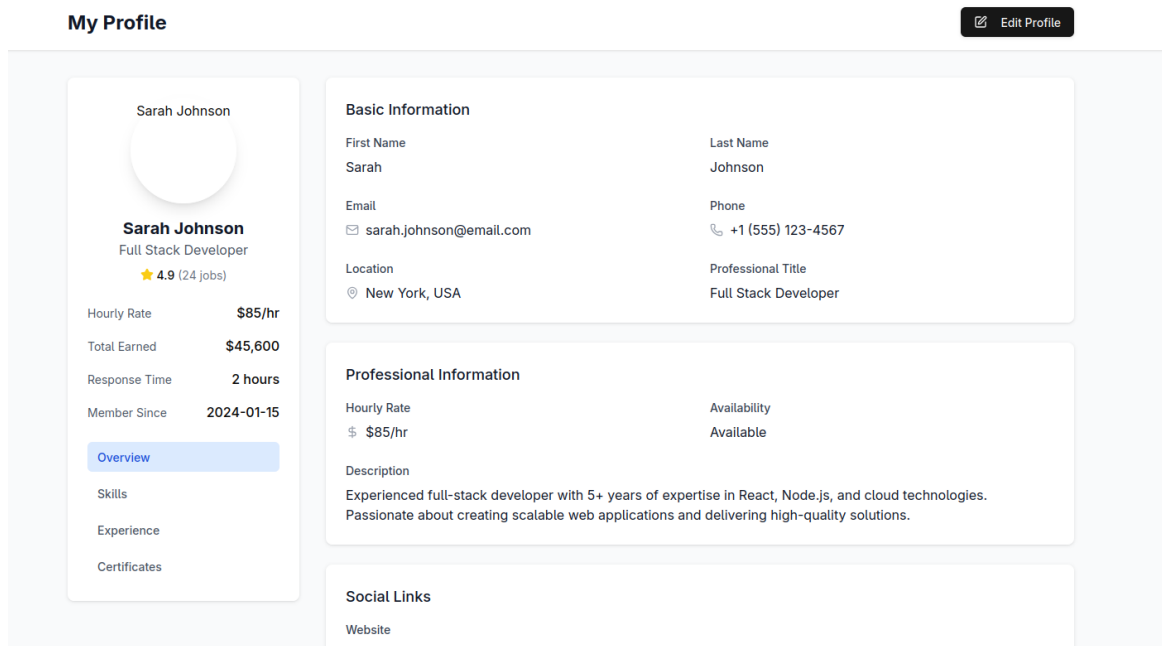


Figure 14: Profile Page

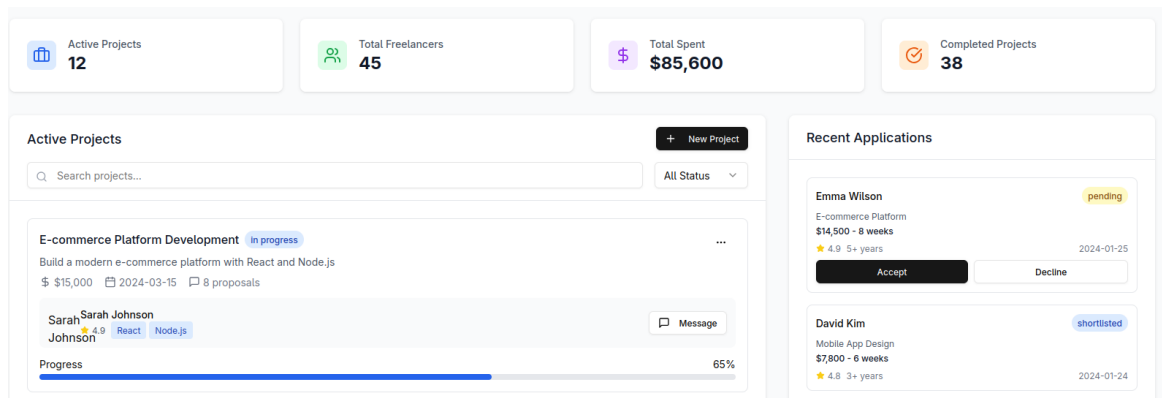


Figure 15: Company Dashboard

Applications

3 total applications

Review and manage freelancer applications for your projects

🔍 Search freelancers...

All Status

All Projects

Sarah **Sarah Johnson**
Johnson ★ 4.9 24 projects
completed

New York,
USA

pending



Applied
2024-01-25

Applied for: E-commerce Platform Development

\$ \$14,500 📅 8 weeks

Skills

React

Node.js

Python

Proposal

I have extensive experience building e-commerce platforms with React and Node.js...



View Profile



Message



Reject

Shortlist



Accept

Figure 16: Application Page

Create a detailed project posting to attract the best talent

Project Details

Project Title *

Project Description *

Describe your project in detail. Include requirements, goals, and any specific preferences...

Category *

Select category

Budget (USD) *

e.g., 5000

Experience Level *

Select experience level

Project Duration *

Select duration

Project Deadline *

mm / dd / yyyy

Required Skills *

Add a skill (e.g., React, Python, Figma)

+

Save as Draft

Post Project

Figure 17: Add Project Page

Browse Freelancers

Find and hire the best freelancers for your projects

4 freelancers available

All Skills

All

Rating

Sarah Johnson

Sarah Johnson

Full Stack Developer

★ 4.9 (127)
📍 New York, USA

Experienced full-stack developer with 6+ years building scalable web applications...

React

Node.js

Python

MongoDB

+1 more

\$85

per hour

24

projects

98%

success rate

Available

Responds in 1 hour

Hire Now

Profile

Message

Figure 18: Browse Freelancer Page

All Status

Join Date

More Filters

5
Total Freelancers

3
Active

1
Pending

1
Suspended

Freelancers (5)

Select All

Sarah Johnson

active

sarah.johnson@email.com

📍 New York, USA

📅 Joined 2024-01-15

★ 4.9

React

Node.js

Python

MongoDB

24 jobs Completed

\$45,600 Total Earned

\$85/hr Rate

View

Edit

Suspend

Delete

Mike Chen

active

mike.chen@email.com

📍 San Francisco, USA

📅 Joined 2024-01-10

★ 4.8

UI/UX Design

Figma

Adobe XD

Prototyping

18 jobs Completed

\$32,400 Total Earned

\$75/hr Rate

View

Edit

Suspend

Delete

Figure 19: User Management Page

3.3.3 Algorithm Details

1. Search & Matching Algorithm

```
def match_freelancers_to_project(project, freelancers):  
    matched_freelancers = []  
    for freelancer in freelancers:  
        # Step 1: Skill Matching Score  
        matched_skills = set(project['skills_required']) &  
set(freelancer['skills'])  
        skill_score = len(matched_skills) /  
len(project['skills_required'])  
        # Step 2: Experience Score (normalized: e.g., 0 to 1)  
        experience_score = min(freelancer['years_experience'] /  
10, 1)  
        # Step 3: Rating Score (out of 5)  
        rating_score = freelancer['rating'] / 5  
        # Step 4: Composite Score  
        total_score = (skill_score * 0.5) + (experience_score *  
0.3) + (rating_score * 0.2)  
        # Step 5: Add to matched list  
        matched_freelancers.append({  
            'freelancer': freelancer,  
            'score': total_score  
        })  
        # Step 6: Sort by score  
        matched_freelancers.sort(key=lambda x: x['score'],  
reverse=True)  
  
    return matched_freelancers
```


2. Proposal Selection Algorithm

```
def select_best_proposals(project, proposals):
    ranked_proposals = []
    for p in proposals:
        # Step 1: Normalize bid fairness (closer to budget = better)
        budget = project['budget']
        bid_score = max(0, 1 - abs(p['bid_amount'] - budget) / budget)

        # Step 2: Normalize experience (max 10 years for score cap)
        experience_score = min(p['freelancer_experience_years'] / 10,
1)

        # Step 3: Normalize rating (out of 5)
        rating_score = p['freelancer_rating'] / 5

        # Step 4: Optional - NLP cover letter score
        # cover_score = get_cover_letter_score(p['cover_letter']) #
Placeholder

        # Step 5: Weighted scoring
        total_score = (
            (bid_score * 0.3) +
            (experience_score * 0.3) +
            (rating_score * 0.4)
        )
        ranked_proposals.append({
            'proposal': p,
            'score': total_score
        })

    # Step 6: Sort proposals by score descending
    ranked_proposals.sort(key=lambda x: x['score'], reverse=True)
    return ranked_proposals
```

Chapter 4: Implementation and Testing

4.1. Implementation:

Implementation mainly refers to the stage of system development. First, we examined, analyzed, and put a system for users in place using all the data we collected. One of the most crucial stages of any project is this one. Coding, testing, installation, documentation, training, and support are typically included in implementation. The system was developed using a variety of tools and technologies, which were covered in the preceding chapter. In simple terms, it is the process of turning a system design specification into functional software.

For NCE, the system was built using modern technologies to ensure efficiency, scalability, and ease of use. The mobile application is designed for both freelancers and companies, making it simple for users to post jobs, apply for gigs, communicate, and complete transactions.

4.1.1. Technologies Used:

- **Django Ninja:**

Django Ninja is a high-performance web framework built on top of Django and FastAPI principles. It was used to develop the back-end APIs of the platform. Django Ninja is well-suited for building fast and secure REST APIs. It handles all the business logic, user authentication, data validation, and routing between the front-end and the PostgreSQL database.

- **Next.js:**

Next.js is a powerful React-based framework used to build fast, scalable, and SEO-friendly web applications. It offers features such as server-side rendering (SSR), static site generation (SSG), and API routes, which make it ideal for building modern, dynamic web interfaces. In this project, Next.js was used for the frontend to design a responsive, interactive, and high-performance user interface for freelancers and companies. Its built-in routing and optimized performance helped streamline the development and enhance the user experience of the platform.

- **PostgreSQL:**

PostgreSQL is an advanced open-source relational database used to store all system data. It handles critical information such as user profiles, job postings, proposals, payments, and reviews. With its support for complex queries and strong data

integrity features, PostgreSQL ensures efficient data storage and retrieval across the system.

This combination of Django Ninja, Next.js and PostgreSQL provides a reliable and scalable system that meets the requirements of freelancers and companies in Nepal. Roles in the system have been clearly defined, ensuring smooth interaction between admins, freelancers, and companies.

4.1.2 Implementation Details of Modules

After the design phase was completed and potential challenges were addressed, the implementation of the freelancing platform began. Below are the key modules developed in the system:

1. Register Form

This module allows new users (freelancers or companies or admin) to sign up for the platform.

- Users provide information such as first name, last name, email, password, and role (freelancer/company/admin).
- The form communicates with the backend (Django Ninja API) to store user data in the PostgreSQL database.
- On successful registration, users are redirected to the login page to access the platform.

2. Login Form

This module allows existing users to log into the system.

- Users input their email and password, which are validated via the backend.
- Based on the role, users are redirected to either the freelancer dashboard, company dashboard, or admin panel.

3. Project Creation (Company Side)

This module allows registered companies to post freelance job opportunities.

- Companies specify project title, description, required skills, budget, and timeline.
- Jobs are stored in the database and visible to freelancers browsing for work.
- Companies can manage their posted projects (edit or delete).

4. Proposal Creation (Freelancer Side)

Freelancers can apply to listed jobs by submitting proposals.

- Proposals include a cover letter, expected fee, and delivery time.
- Proposals are stored with references to the corresponding job and freelancer ID.
- Companies review proposals and can accept or decline them.

5. Admin Module

Admins manage the backend operations of the platform.

- Admins can view, verify, or delete user accounts, job posts, and proposals.
- Admin dashboard includes analytics on users, jobs, and platform activity.
- Admins ensure proper content moderation and handle disputes if necessary.

6. Customer Module (Freelancer and Company Panels)

These are user-specific dashboards depending on the registered role.

- Freelancers can update their profiles, view submitted proposals, track jobs, and check earnings.
- Companies can view proposals, manage projects, communicate with freelancers, and leave feedback.
- Dashboards provide activity summaries, notifications, and options for account settings.

4.2. Testing

Based on the software requirement specification document, testing was performed to investigate and validate the behavior of a fully integrated software product. Before deploying an application or website, it must be thoroughly tested. As a result, this application's test cases were written. Some of the types of testing that we did are described below.

4.2.1. Test cases for unit testing

Unit testing emphasizes the verification effort on the smallest unit of software design i.e., a software component or module. Unit testing is performed in parallel with the coding phase. Unit testing tests units or modules not the whole software. I have tested each view/module of the application individually. As the modules were built up testing was

carried out simultaneously, tracking out each and every kind of input and checking the corresponding output until module is working correctly.

Table 1: Test Case for Admin's Login

Test Case Id	Test Case Description	Test Data	Expected result	Actual result	Pass/Fail
1	Enter a wrong email	Email: adminsa Password: ad123456 Role: Admin	Display message: "Email is invalid"	As expected	Pass
2	Enter a wrong password	Email: admin@gmail.com Password: ads Role: Admin	Display message: "Password is invalid"	As expected	Pass
3	Enter a wrong role	Email: admin@gmail.com Password: adsfniss Role: student	Display message: "Role is invalid"	As expected	Pass
4	Enter correct email, password and role	Email: admin@gmail.com Password: admin Role: Admin	Logged into Admin Page	As expected	Pass

Table 2: Test Case for Freelancer's Login

Test Case Id	Test Case Description	Test Data	Expected result	Actual result	Pass/Fail
1	Enter a wrong email	Email: freelancersaa Password: fre123456 Role: Freelancer	Display message: "Email is invalid"	As expected	Pass
2	Enter a wrong password	Email: Freelancer@gmail.com Password: ads Role: Freelancer	Display message: "Password is invalid"	As expected	Pass
3	Enter a wrong role	Email: Freelancer@gmail.com Password: fre123456 Role: Faculty	Display message: "Role is invalid"	As expected	Pass
4	Enter correct email, password and role	Email: Freelancer@gmail.com Password: fre123456 Role: Freelancer	Logged into Freelancer Page	As expected	Pass

Table 3: Test Case for Company's Login

Test Case Id	Test Case Description	Test Data	Expected result	Actual result	Pass/Fail
1	Enter a wrong email	Email: companyaa Password: comp3456 Role: Company	Display message: "Email is invalid"	As expected	Pass
2	Enter a wrong password	Email: company@gmail.com Password: ads Role: Company	Display message: "Password is invalid"	As expected	Pass
3	Enter a wrong role	Email: company@gmail.com Password: comp3456 Role: Student	Display message: "Role is invalid"	As expected	Pass
4	Enter correct email, password and role	Email: company@gmail.com Password: comp3456 Role: Company	Logged into Company Page	As expected	Pass

Table 4: Add Project Test Case

Test Id	Test Case	Test Data	Expected Result	Actual Result	Pass/Fail
1	Add project	Add project	Project added successfully	As expected,	Pass

Table 5: Delete Project Test Case

Test Id	Test Case	Test Data	Expected Result	Actual Result	Pass/Fail
1	Delete project	Delete project from website	Project delete successfully	As expected,	Pass

Table 6: Update Project Test Case

Test Id	Test Case	Test Data	Expected Result	Actual Result	Pass/Fail
1	Update project	Update project	Project updated	As expected,	Pass

Table 7: Register User Test Case

Test Id	Test Case	Test Data	Expected Result	Actual Result	Pass/Fail
1	Register with some empty field	First name: Demo Last name: Demo Email: <u>demo@gmail.com</u> Password: NULL Role: NULL	Please fill out empty field	As expected,	Pass
2	Enter already used email	Email: demo@gmail.com	Email already exist	As expected,	Pass
3	Enter an invalid email	Email: demogmail.com	Invalid email	As expected,	Pass

4	Enter unmatched password in retype password	Password: demo12345 Password: 123456	Password doesn't match	As expected,	Pass
5	Register with every field filled	First name: Demo Last name: Demo Email: <u>demo@gmail.com</u> Password: san45678 Role: Freelancer	User should be registered	As expected,	Pass

Chapter 5: Conclusion and Future Recommendations

5.1. Conclusion

In conclusion, this project – a Freelancing Platform – is designed to be flexible, user-friendly, and accessible. Most of the core requirements have been successfully implemented, and the remaining minor enhancements can be added with minimal extension. The system is developed using Next.js for the frontend, Django Ninja for the backend, and PostgreSQL for the database, making it a robust and scalable application suitable for real-world deployment.

This freelancing platform serves as a digital bridge between companies and freelancers in Nepal, providing a seamless experience for both parties. Freelancers can register, create detailed profiles, showcase their skills and portfolios, and apply for job listings. On the other hand, companies can post job requirements, view freelancer profiles, and hire professionals based on reviews and experience. Admins have the ability to manage users, monitor job activity, and maintain the integrity of the platform. Overall, this platform promotes digital employment and aims to support the growing freelance economy in Nepal.

5.2. Outcome

Upon completion of the project, the Freelancing Platform system has been successfully developed with all key functionalities working as intended. The system supports role-based access for Admins, companies, and Freelancers.

- **Freelancers** can sign up, create their profile, update skills, browse job listings, submit proposals, and get hired.
- **Companies** can register, post jobs, manage incoming proposals, and hire suitable freelancers for their projects.
- **Admins** can manage user accounts, review flagged content or disputes, and oversee overall platform activity.

The admin dashboard displays statistics like the number of registered freelancers, companies, jobs posted, and completed contracts, providing a complete overview of system operations. The system ensures transparency and accountability through profile ratings and a review system.

5.3. Future Recommendations

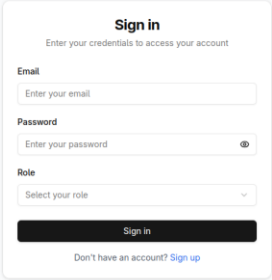
The project has been structured to allow easy updates and feature expansions in future versions. While the current implementation covers all the essential features, there are several enhancements that could significantly improve the platform:

- **Mobile Application:** Develop a dedicated mobile app for Android and iOS to enhance accessibility and user engagement.
- **Multilingual Support:** Implement multiple language options to make the platform inclusive for users from diverse regions in Nepal.
- **Dispute Resolution System:** Introduce a structured conflict resolution mechanism to handle disputes between freelancers and companies.
- **AI Job Matching:** Use AI/ML algorithms to suggest job opportunities to freelancers based on their skills and past performance.

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Appendices



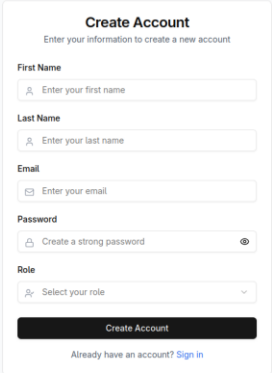
Sign in
Enter your credentials to access your account

Email

Password

Role

Don't have an account? [Sign up](#)



Create Account
Enter your information to create a new account

First Name

Last Name

Email

Password

Role

Already have an account? [Sign in](#)

NCE(Nepal Freelance)

Dashboard

View Applications

Add Project

View Projects

Browse Freelancers

Dashboard

Manage your projects and find the best freelancers

Active Projects12

Total Freelancers45

Total Spent\$85,600

Completed Projects38

Active Projects

Search projects...

All Status

E-commerce Platform Development

in progress

Build a modern e-commerce platform with React and Node.js

\$ \$15,000 2024-03-15 8 proposals

Sarah Johnson

4.9 React Node.js

Message

Progress65%

Mobile App UI/UX Design

review

Design modern and intuitive mobile app interface

\$ \$8,000 2024-02-28 12 proposals

Mike Chen

4.8 UI/UX Figma

Message

Progress90%

Content Marketing Strategy

pending

Recent Applications

Emma Wilson

E-commerce Platform

\$14,500 - 8 weeks

4.9 5+ years

2024-01-25

AcceptDecline

David Kim

Mobile App Design

\$7,800 - 6 weeks

4.8 3+ years

2024-01-24

View All Applications

Quick Actions

Post New Project

Browse Freelancers

View Applications

View Reports

NCE(Nepal Freelance)

Add Project

View Applications

Add Project

View Projects

Browse Freelancers

Add Project

Manage your projects and find the best freelancers

Post a New Project

Create a detailed project posting to attract the best freelancers

Project Details

Project Title *

e.g., Build a modern e-commerce website

Project Description *

Describe your project in detail. Include requirements, goals, and any specific preferences...

Category *

Select category

Budget (INR) *

e.g., 5000

Experience Level *

Select experience level

Project Duration *

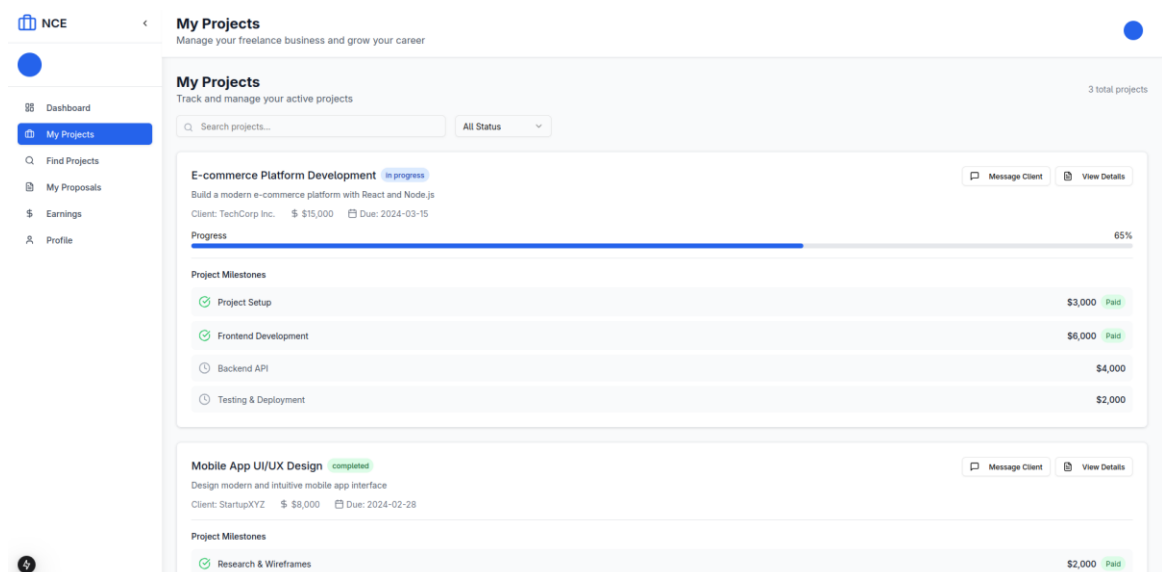
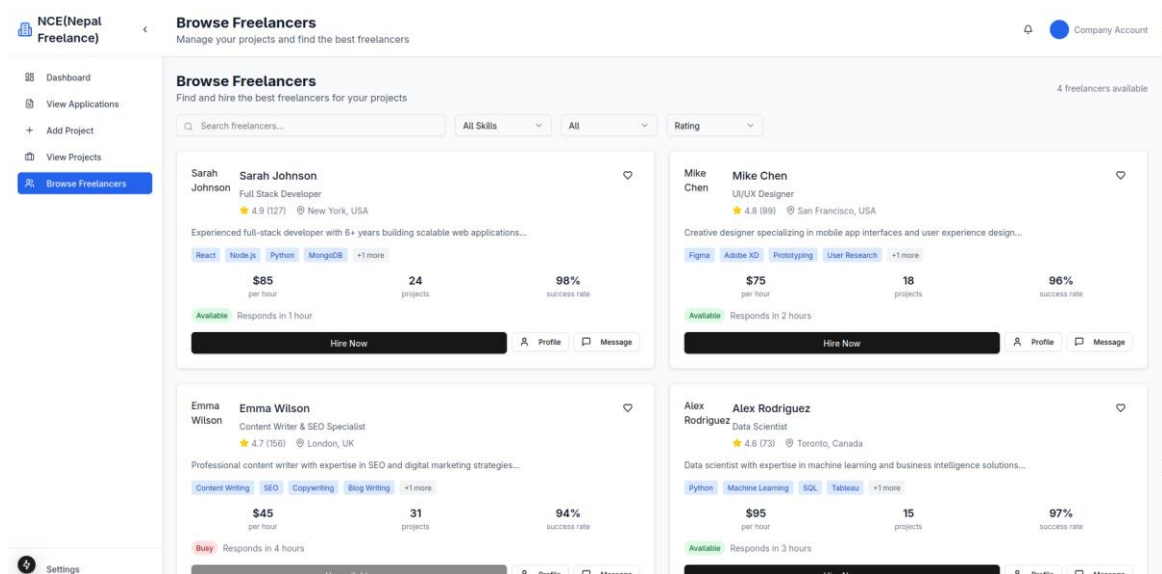
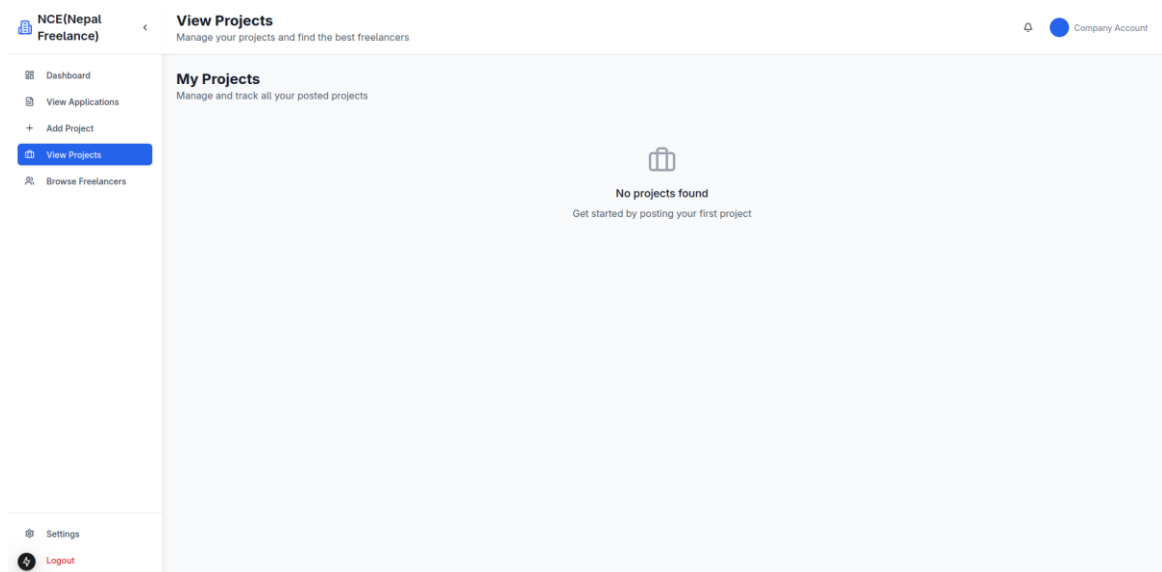
Select duration

Project Deadline *

mm / dd / yyyy

Required Skills *

Add a skill (e.g., React, Python, Figma)



NCE

Dashboard

My Projects

Find Projects

My Proposals

Earnings

Profile

Find Projects

Manage your freelance business and grow your career

Find Projects

Discover new opportunities that match your skills

3 projects available

Search projects...

All Categories

All Budgets

Newest

React Native Mobile App Development

We need an experienced React Native developer to build a cross-platform mobile app for our e-commerce business. The app should include user authentication, product catalog, shopping cart, and payment integration.

\$ 8,000 - \$12,000

2-3 months

San Francisco, USA

React Native

JavaScript

API Integration

Mobile Development

T

TechStart Solutions

Verified

Rating: 4.8/5

12 proposals

Posted 2024-01-25

Experience: Intermediate

Category: Mobile Development

View Details

Send Proposal

UI/UX Design for SaaS Dashboard

Looking for a talented UI/UX designer to create a modern, intuitive dashboard for our SaaS platform. The design should be clean, professional, and user-friendly with a focus on data visualization.

\$ 5,000 - \$8,000

4-6 weeks

New York, USA

UI/UX Design

Figma

Prototyping

User Research

D

DataFlow Inc.

Verified

Rating: 4.9/5

8 proposals

Posted 2024-01-24

Experience: Expert

Category: Design

View Details

Send Proposal

NCE

Dashboard

My Projects

Find Projects

My Proposals

Earnings

Profile

My Proposals

Manage your freelance business and grow your career

My Proposals

Track and manage your project proposals

4 total proposals

Search proposals...

All Status

React Native Mobile App Development

pending

Client: TechStart Solutions

\$ 10,000

10 weeks

Submitted: 2024-01-25

Cover Letter

I have 5+ years of experience in React Native development and have built 20+ mobile apps...

Edit

View

UI/UX Design for SaaS Dashboard

accepted

Client: DataFlow Inc.

\$ 6,500

5 weeks

Submitted: 2024-01-24

Cover Letter

I specialize in SaaS dashboard design and have worked with similar companies...

View

Client Response

Great portfolio! We'd like to move forward with your proposal.

Message Client

38

